

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 1 (Generated practice paper - Set 34)**

Assessed: The physical environment, Energy resources, Pollution, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

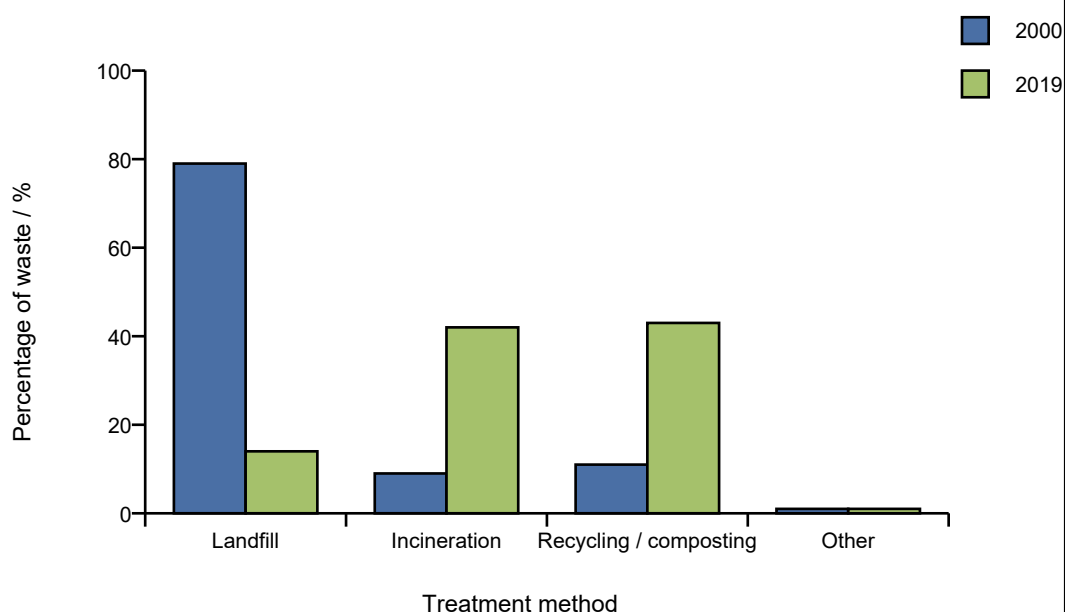
Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1

Figure 1 shows how municipal solid waste was treated in the UK in 2000 and 2019.

Figure 1



0 1 . 1

Describe the changes shown in Figure 1.

[2 marks]

0 1 . 2

Explain **three** environmental disadvantages of landfill that led to the change.

[3 marks]

1

2

3

0 1 . 3

Describe **four** features of a well-managed landfill site.

[4 marks]

1

2

3

4

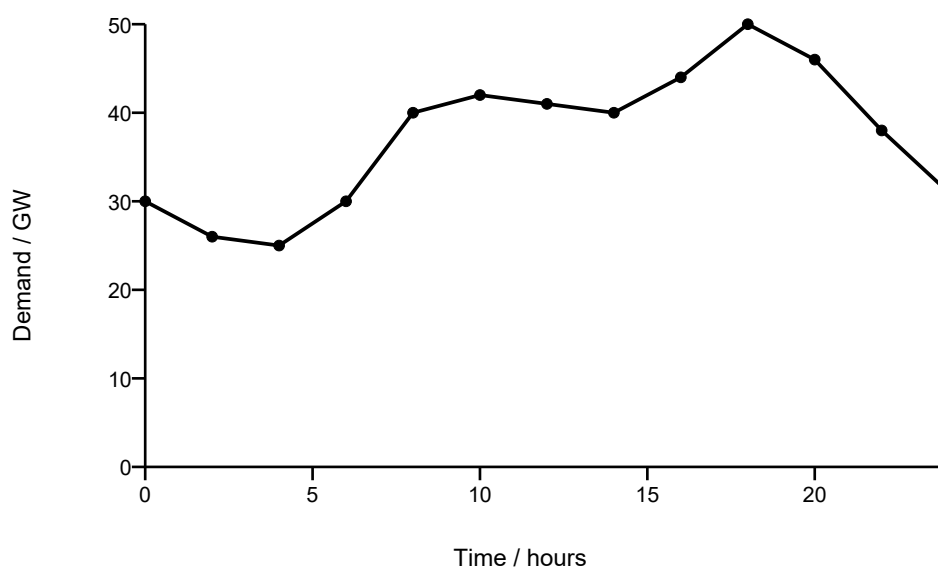
0 1 . **4** Suggest why incineration has increased more than recycling for some waste types.

[1 mark]

10

0 2 Figure 1 shows the demand for electricity in the UK over a 24-hour period on a weekday in winter.

Figure 1



0 2 . **1** Describe the pattern of demand shown in Figure 1.

[2 marks]

0 2 . 2

Explain the causes of the fluctuations shown in Figure 1 and give **two** other causes of fluctuations in electricity demand.

[4 marks]

0 2 . 3

Explain why base-load power stations (nuclear, large coal) produce a surplus of electricity at night.

[2 marks]

0 2 . 4

Give **two** causes of fluctuations in energy **supply**.

[2 marks]

1

2

10

03 . 1

Explain how the control of emission **location** and emission **timing** can reduce the severity of pollution. Use examples.

[6 marks]

Emission location

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Emission timing

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03 . 2

Explain the **polluter pays** principle and suggest why it encourages pollution prevention.

[2 marks]

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03 . 3

Explain the **precautionary principle** as applied to the release of a new chemical.

[2 marks]

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0 4

Table 1 shows information about two copper deposits, X and Y, being assessed for mining. The current cut-off ore grade for copper is 0.4%.

Table 1

Feature	Deposit X	Deposit Y
Ore grade / % copper	0.9	0.35
Depth of deposit / m	350	40
Overburden	Hard igneous rock	Loose glacial sediment
Distance to nearest railway / km	120	8
Annual rainfall / mm	2 100	500
Chemical form	Copper sulfide	Copper sulfide

0 4

1

Explain what is meant by the **cut-off ore grade** and state which deposit is currently included in the copper reserves.

[2 marks]

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0 4

2

Use Table 1 to discuss the factors that affect the viability of mining each deposit.

[6 marks]

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 0 4 . 3 State **two** changes that would decrease the cut-off ore grade. [2 marks]

1

2

10

0 5 A student estimated the population of ground beetles in a field using the Lincoln Index. 60 beetles were caught in pitfall traps, marked with a spot of paint and released. Three days later 45 beetles were caught, of which 9 were marked.

0 5 . 1 Calculate the estimated population size. [2 marks]

Answer beetles

0 5 . 2 State **four** assumptions of the Lincoln Index method. [4 marks]

1

2

3

4

0 5 . **3** Explain how the estimate would be affected if the paint made the beetles more likely to be eaten by birds.

[2 marks]

0 5 . **4** Suggest why the second sample was taken after three days rather than the next day or after a month.

[2 marks]

10

Table 1 shows features of oil tanker design and operation that reduce the risk of oil pollution.

Complete Table 1 by explaining how each feature reduces oil pollution.

Do not write outside the box

Table 1

Feature	How it reduces oil pollution	
Double hull		
Inert gas system		
Separate oil and ballast tanks		
Twin engines / rudders		
Recirculation of oil during unloading (crude oil washing)		

Explain how each of the following reduces the risk of tanker accidents.

[2 marks]

Offshore shipping routes

GPS / Automatic Identification System

06 . 3

Describe how a **bund wall** prevents oil pollution from a storage tank.

[2 marks]

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06 . 4

Explain the purpose of an oil interceptor on the drainage system of a car park.

[1 mark]

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10

07

Table 1 shows the water quality required for four different uses.

07 . 1

Complete Table 1 by giving the quality requirement for each water use and the reason for it.

[4 marks]

Table 1

Water use	Quality requirement	Reason	
Potable (public) supply			
Spray irrigation			
Power station condenser cooling			
Industrial boiler water			

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07 . 2

Explain why aeration is used in the treatment of water from anaerobic sources.

[2 marks]

07 . 3

Explain how flocculation removes clay particles that sedimentation cannot.

[2 marks]

07 . 4

Give **two** methods used to sterilise treated water.

[2 marks]

1

2

10

08 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

- A High sand content
B High clay content
C Crumb peds
D High pH

☐
☐
☐
☐

08 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

08 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

5

09 . 1

Complete Table 1 to show one method of controlling each source of railway noise.

[4 marks]

Table 1

Source of railway noise	Control method	
Wheel vibration on the track		
Pantograph turbulence		
Wheel squeal on corners		
Braking squeal		

09 . 2

Explain why road traffic noise affects more people than any other form of noise pollution.

[1 mark]

09 . 3

Describe **three** ways of reducing road traffic noise reaching residents.

[3 marks]

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2

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3

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09 . 4

Suggest how noise from pile-driving at sea for wind turbine foundations can be reduced to protect cetaceans.

[2 marks]

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10

10 . 1

Evaluate the extent to which new technologies could allow fossil fuels and nuclear fission to continue to supply a large proportion of the world's energy sustainably.

[9 marks]

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10 . **2**

State the approximate percentage of natural gas in a reservoir that can be recovered using its own pressure.

[1 mark]

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10

Write an essay on one of the following topics.

$$\boxed{1} \boxed{1} . \boxed{1}$$

Discuss the difficulties in monitoring and predicting global climate change and evaluate the methods used to overcome them.

[25 marks]

OR

$$\boxed{1} \boxed{1} . \boxed{2}$$

Discuss how knowledge of the properties of energy resources can be used to reduce the severity of their impact on the environment.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There is no handwriting or other markings on the paper.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

Do not write outside the box

END OF QUESTIONS